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Electromagnetic

Assignment 3

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1 Problem 1

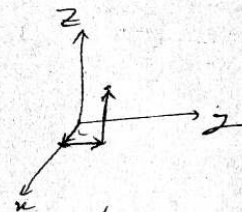
$$\nabla \times \vec{E} = 0$$

(ف)

$$\begin{aligned} \nabla \times \vec{E} &= \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{i} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) \hat{j} \\ &\quad + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) \hat{k} \\ &= (2y - Ay) \hat{i} + 0 \hat{j} + 0 \hat{k} \end{aligned}$$

$$A = 2$$

$$\int_C \vec{E} \cdot d\vec{l} = -V_p$$



به دلیل اینکه $\nabla \times \vec{E} = 0$ ، انتگرال از هر مسیری منتهی به یک نقطه برابر است.

$$\int_0^{x_0} (\vec{E} \cdot \hat{x}) dx + \int_0^{y_0} (\vec{E} \cdot \hat{y}) dy + \int_0^{z_0} (\vec{E} \cdot \hat{z}) dz = -V_p$$

$$(\vec{E} \cdot \hat{x})_{(x=x_0, y=y_0, z=0)} = x_0 \quad (\vec{E} \cdot \hat{y})_{(x=x_0, y=y_0, z=0)} = y_0$$

$$(\vec{E} \cdot \hat{z})_{(x=x_0, y=y_0, z=z_0)} = y_0^2 - z_0^2$$

$$V_{(k=1, j=2, z=1)} = \frac{1}{2} + 2 + 4 - \frac{1}{3} = 6 + \frac{1}{6}$$

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho$$

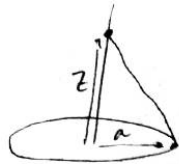
$$\nabla \cdot \mathbf{E} = 1 + (2z+1) - 2z = 2 \Rightarrow \rho_{(x,y,z)} = 2\epsilon_0$$

$$\int \nabla \cdot \mathbf{E} \, d\tau = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{2\epsilon_0 z'}{\epsilon_0} = 2 \quad (1)$$

2 Problem 2

سوال 2

الف)



$$V_{(0,0,z)} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r}$$

$$dq = \lambda a d\theta$$

$$V_{(0,0,z)} = \frac{a}{4\pi\epsilon_0} \int_0^{2\pi} \frac{\lambda d\theta}{\sqrt{a^2 + z^2}} = 0$$

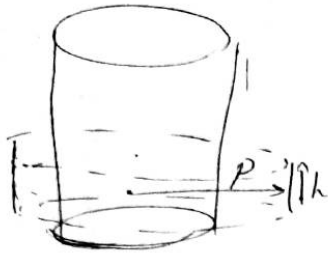
ب) میدان در راستای x, z و E موازی محور x است.

$$\vec{E} = \frac{\hat{j}}{4\pi\epsilon_0} \int_{-\pi}^{\pi} \frac{a \lambda d\theta}{(a^2 + z^2)^{3/2}} (-a \sin\theta)$$

$$= -\frac{\hat{j}}{4\pi\epsilon_0} \frac{4a^2 \lambda}{(a^2 + z^2)^{3/2}} \quad , \quad Q = \pi a \lambda$$

$$= -\frac{\hat{j}}{\pi \epsilon_0} \frac{aQ}{(a^2 + z^2)^{3/2}}$$

3 Problem 3



$$\int \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q$$

$$P > a:$$

$$Q = h \times \pi a^2 P_0$$

$$|\vec{E}_0| \text{ (or } K) = \frac{1}{\epsilon_0} K a P_0$$

$$\vec{E}(r) = \frac{a^2 P_0}{2\epsilon_0} \frac{1}{r} \hat{r}$$

$$P < a$$

$$Q = h \times \pi P_0$$

$$\vec{E}(r) = \frac{P_0}{2\epsilon_0} r \hat{r}$$

ب) به دلیل آنکه برهم نمی‌تابند

$$\vec{E} = \frac{P_0}{2\epsilon_0} \vec{R}_1 - \frac{P_0}{2\epsilon_0} (\vec{R}_1 - \vec{J})$$

$$= \cancel{\frac{P_0}{2\epsilon_0}} = \frac{P_0}{2\epsilon_0} \vec{J} = \frac{P_0}{2\epsilon_0} d \hat{x}$$

4 Problem 4

از اصل برهم ننی استفاده می کنیم.

بست آردن میدان صند بی نهایت:

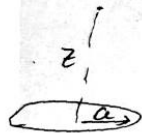


$$2|E_{\text{int}}|A = \frac{P_s}{\epsilon_0} A$$

$$\rightarrow \vec{E}_{\text{int}} = \frac{P_s}{2\epsilon_0} \hat{z}$$

میدان دیک به شایع:

به دلیل تشار در راستای z نقطه میدان صند دارد:



$$\vec{E} = \hat{k} \int \frac{P_s \pi r dr}{4\pi\epsilon_0 (r^2 + z^2)^{3/2}} (z)$$

$$\vec{E}_2 = \hat{k} \frac{2\pi P_s z}{4\pi\epsilon_0} \int_0^a \frac{r dr}{(r^2 + z^2)^{3/2}}$$

$$\left(\frac{1}{z} - \frac{1}{\sqrt{a^2 + z^2}} \right)$$

$$\vec{E} = \vec{E}_1 - \vec{E}_2 = \frac{P_s}{2\epsilon_0} \frac{z}{\sqrt{a^2 + z^2}} \hat{k}$$